

1 Definitions

$$\text{Form A:} \quad \bar{a} = 1 - e^{-\frac{b}{p+1}(\bar{w}_d - \bar{w}_s)^{p+1}} \quad (1)$$

$$\bar{a}'(\bar{w}_d - \bar{w}_s, b, p) = b \cdot (\bar{w}_d - \bar{w}_s)^p \cdot e^{-\frac{b}{p+1}(\bar{w}_d - \bar{w}_s)^{p+1}} \quad (2)$$

$$\bar{a}'(\bar{w}_d - \hat{w}_s, \hat{b}, \hat{p}) = \hat{b} \cdot (\bar{w}_d - \hat{w}_s)^{\hat{p}} \cdot e^{-\frac{\hat{b}}{\hat{p}+1}(\bar{w}_d - \hat{w}_s)^{\hat{p}+1}} \quad (3)$$

$$\text{Form B:} \quad \bar{a} = 1 - \left[\frac{b}{[\bar{w}_d - \bar{w}_s + b]} \right]^p \quad (4)$$

$$\bar{a}'(\bar{w}_d - \bar{w}_s, b, p) = p b^p \cdot [\bar{w}_d - \bar{w}_s + b]^{-p-1} \quad (5)$$

$$\bar{a}'(\bar{w}_d - \hat{w}_s, \hat{b}, \hat{p}) = \hat{p} \hat{b}^{\hat{p}} \cdot [\bar{w}_d - \hat{w}_s + \hat{b}]^{-\hat{p}-1} \quad (6)$$

$$\bar{w}_d - \bar{w}_s > 0, \quad \bar{w}_d - \hat{w}_s > 0 \quad (7)$$

$$b > 0, \quad \hat{b} > 0, \quad p > -1, \quad \hat{p} > -1 \quad (8)$$

Both forms place contact at $\bar{w}_d = \bar{w}_s$:

$$\bar{a} \Big|_{\bar{w}_d = \bar{w}_s} = 1 - e^0 = 0 \quad (\text{Form A}) \quad (9)$$

$$\bar{a} \Big|_{\bar{w}_d = \bar{w}_s} = 1 - \left[\frac{b}{b} \right]^p = 1 - 1^p = 0 \quad (\text{Form B, any } p) \quad (10)$$

2 Parameter errors

$$e_b = b - \hat{b}, \quad e_p = p - \hat{p}, \quad e_{w_s} = \bar{w}_s - \hat{w}_s, \quad e_{\bar{w}_d} \equiv 0 \quad (11)$$

$$\mathbf{e}_A = \begin{bmatrix} e_b & e_{w_s} & e_p \end{bmatrix}^\top, \quad \mathbf{e}_B = \begin{bmatrix} e_b & e_p & e_{w_s} \end{bmatrix}^\top \quad (12)$$

3 First-order expansion

$$\begin{aligned} \bar{a}'(\bar{w}_d - \bar{w}_s, b, p) - \bar{a}'(\bar{w}_d - \hat{w}_s, \hat{b}, \hat{p}) \\ = J_b(\bar{w}_d - \hat{w}_s, \hat{b}, \hat{p}) e_b + J_{w_s}(\bar{w}_d - \hat{w}_s, \hat{b}, \hat{p}) e_{w_s} + J_p(\bar{w}_d - \hat{w}_s, \hat{b}, \hat{p}) e_p + \mathcal{O}(\|\mathbf{e}_A\|^2) \end{aligned} \quad (13)$$

$$\begin{aligned}
& \bar{a}'(\bar{w}_d - \bar{w}_s, b, p) - \bar{a}'(\bar{w}_d - \hat{w}_s, \hat{b}, \hat{p}) \\
& = J_b(\bar{w}_d - \hat{w}_s, \hat{b}, \hat{p}) e_b + J_p(\bar{w}_d - \hat{w}_s, \hat{b}, \hat{p}) e_p + J_{w_s}(\bar{w}_d - \hat{w}_s, \hat{b}, \hat{p}) e_{w_s} + \mathcal{O}(\|e_B\|^2) \quad (14)
\end{aligned}$$

$$\frac{\partial \bar{a}'}{\partial x} = \bar{a}' \cdot \frac{\partial \ln \bar{a}'}{\partial x} \quad (15)$$

4 Form A: derivatives of the logarithm

$$\ln \bar{a}'(\bar{w}_d - \bar{w}_s, b, p) = \ln b + p \ln(\bar{w}_d - \bar{w}_s) - \frac{b}{p+1} (\bar{w}_d - \bar{w}_s)^{p+1} \quad (16)$$

$$\begin{aligned} \frac{\partial \ln \bar{a}'}{\partial b} &= \frac{1}{b} - \frac{\partial}{\partial b} \left[\frac{b}{p+1} (\bar{w}_d - \bar{w}_s)^{p+1} \right] \\ &= \frac{1}{b} - \frac{1}{p+1} (\bar{w}_d - \bar{w}_s)^{p+1} \end{aligned} \quad (17)$$

$$\begin{aligned} \frac{\partial \ln \bar{a}'}{\partial \bar{w}_s} &= p \cdot \frac{-1}{(\bar{w}_d - \bar{w}_s)} - \frac{b}{p+1} (p+1) (\bar{w}_d - \bar{w}_s)^p \cdot (-1) \\ &= -\frac{p}{(\bar{w}_d - \bar{w}_s)} + b (\bar{w}_d - \bar{w}_s)^p \end{aligned} \quad (18)$$

$$\frac{\partial}{\partial p} (\bar{w}_d - \bar{w}_s)^{p+1} = \frac{\partial}{\partial p} e^{(p+1) \ln(\bar{w}_d - \bar{w}_s)} = (\bar{w}_d - \bar{w}_s)^{p+1} \ln(\bar{w}_d - \bar{w}_s) \quad (19)$$

$$\begin{aligned} \frac{\partial}{\partial p} \left[\frac{b}{p+1} (\bar{w}_d - \bar{w}_s)^{p+1} \right] &= b \frac{(\bar{w}_d - \bar{w}_s)^{p+1} \ln(\bar{w}_d - \bar{w}_s) (p+1) - (\bar{w}_d - \bar{w}_s)^{p+1}}{(p+1)^2} \\ &= \frac{b}{p+1} (\bar{w}_d - \bar{w}_s)^{p+1} \left[\ln(\bar{w}_d - \bar{w}_s) - \frac{1}{p+1} \right] \end{aligned} \quad (20)$$

$$\begin{aligned} \frac{\partial \ln \bar{a}'}{\partial p} &= \ln(\bar{w}_d - \bar{w}_s) - \frac{b}{p+1} (\bar{w}_d - \bar{w}_s)^{p+1} \left[\ln(\bar{w}_d - \bar{w}_s) - \frac{1}{p+1} \right] \\ &= \left[1 - \frac{b}{p+1} (\bar{w}_d - \bar{w}_s)^{p+1} \right] \ln(\bar{w}_d - \bar{w}_s) + \frac{b}{(p+1)^2} (\bar{w}_d - \bar{w}_s)^{p+1} \end{aligned} \quad (21)$$

5 Form A: Jacobians

$$J_b(\bar{w}_d - \hat{w}_s, \hat{b}, \hat{p}) = \bar{a}'(\bar{w}_d - \hat{w}_s, \hat{b}, \hat{p}) \cdot \left[\frac{1}{\hat{b}} - \frac{1}{\hat{p}+1} (\bar{w}_d - \hat{w}_s)^{\hat{p}+1} \right] \quad (22)$$

$$J_{w_s}(\bar{w}_d - \hat{w}_s, \hat{b}, \hat{p}) = \bar{a}'(\bar{w}_d - \hat{w}_s, \hat{b}, \hat{p}) \cdot \left[\hat{b} (\bar{w}_d - \hat{w}_s)^{\hat{p}} - \frac{\hat{p}}{(\bar{w}_d - \hat{w}_s)} \right] \quad (23)$$

$$\begin{aligned} J_p(\bar{w}_d - \hat{w}_s, \hat{b}, \hat{p}) &= \bar{a}'(\bar{w}_d - \hat{w}_s, \hat{b}, \hat{p}) \cdot \left[\left(1 - \frac{\hat{b}}{\hat{p}+1} (\bar{w}_d - \hat{w}_s)^{\hat{p}+1} \right) \ln(\bar{w}_d - \hat{w}_s) \right. \\ &\quad \left. + \frac{\hat{b}}{(\hat{p}+1)^2} (\bar{w}_d - \hat{w}_s)^{\hat{p}+1} \right] \end{aligned} \quad (24)$$

6 Form B: derivatives of the logarithm

b is a length, so it enters the bracket additively:

$$\frac{\partial}{\partial b} [\bar{w}_d - \bar{w}_s + b] = 1, \quad \frac{\partial}{\partial p} [\bar{w}_d - \bar{w}_s + b] = 0, \quad \frac{\partial}{\partial \bar{w}_s} [\bar{w}_d - \bar{w}_s + b] = -1 \quad (25)$$

$$\ln \bar{a}'(\bar{w}_d - \bar{w}_s, b, p) = \ln p + p \ln b - (p+1) \ln [\bar{w}_d - \bar{w}_s + b] \quad (26)$$

$$\frac{\partial \ln \bar{a}'}{\partial b} = \frac{p}{b} - \frac{p+1}{[\bar{w}_d - \bar{w}_s + b]} \quad (27)$$

$$\frac{\partial \ln \bar{a}'}{\partial p} = \frac{1}{p} + \ln b - \ln [\bar{w}_d - \bar{w}_s + b] = \frac{1}{p} + \ln \frac{b}{[\bar{w}_d - \bar{w}_s + b]} \quad (28)$$

$$\frac{\partial \ln \bar{a}'}{\partial \bar{w}_s} = -\frac{p+1}{[\bar{w}_d - \bar{w}_s + b]} \cdot (-1) = \frac{p+1}{[\bar{w}_d - \bar{w}_s + b]} \quad (29)$$

7 Form B: Jacobians

$$J_b(\bar{w}_d - \hat{w}_s, \hat{b}, \hat{p}) = \bar{a}'(\bar{w}_d - \hat{w}_s, \hat{b}, \hat{p}) \cdot \left[\frac{\hat{p}}{\hat{b}} - \frac{\hat{p}+1}{[\bar{w}_d - \hat{w}_s + \hat{b}]} \right] \quad (30)$$

$$J_p(\bar{w}_d - \hat{w}_s, \hat{b}, \hat{p}) = \bar{a}'(\bar{w}_d - \hat{w}_s, \hat{b}, \hat{p}) \cdot \left[\frac{1}{\hat{p}} + \ln \frac{\hat{b}}{[\bar{w}_d - \hat{w}_s + \hat{b}]} \right] \quad (31)$$

$$J_{\bar{w}_s}(\bar{w}_d - \hat{w}_s, \hat{b}, \hat{p}) = \bar{a}'(\bar{w}_d - \hat{w}_s, \hat{b}, \hat{p}) \cdot \frac{\hat{p}+1}{[\bar{w}_d - \hat{w}_s + \hat{b}]} \quad (32)$$

8 Form B: Jacobians of the gain itself

$$\frac{\partial \bar{a}}{\partial b} = -p \left[\frac{b}{[\bar{w}_d - \bar{w}_s + b]} \right]^p \frac{\bar{w}_d - \bar{w}_s}{b [\bar{w}_d - \bar{w}_s + b]} \quad (33)$$

$$\frac{\partial \bar{a}}{\partial p} = \left[\frac{b}{[\bar{w}_d - \bar{w}_s + b]} \right]^p \ln \frac{[\bar{w}_d - \bar{w}_s + b]}{b} \quad (34)$$

$$\frac{\partial \bar{a}}{\partial \bar{w}_s} = -p \left[\frac{b}{[\bar{w}_d - \bar{w}_s + b]} \right]^p \frac{1}{[\bar{w}_d - \bar{w}_s + b]} = -\bar{a}'(\bar{w}_d - \bar{w}_s, b, p) \quad (35)$$

9 Form B at $p = 1$

$$\bar{a} = \frac{\bar{w}_d - \bar{w}_s}{[\bar{w}_d - \bar{w}_s + b]}, \quad \bar{a}' = \frac{b}{[\bar{w}_d - \bar{w}_s + b]^2} \quad (36)$$

$$\frac{\partial \bar{a}'}{\partial b} = \frac{(\bar{w}_d - \bar{w}_s) - b}{[\bar{w}_d - \bar{w}_s + b]^3}, \quad \frac{\partial \bar{a}'}{\partial \bar{w}_s} = \frac{2b}{[\bar{w}_d - \bar{w}_s + b]^3} \quad (37)$$

10 Evaluation cost

	$\bar{a}'$	transcendentals required
Form A	$b (\bar{w}_d - \bar{w}_s)^p e^{-\frac{b}{p+1}(\bar{w}_d - \bar{w}_s)^{p+1}}$	pow, exp, log
Form B	$p b^p [\bar{w}_d - \bar{w}_s + b]^{-p-1}$	pow, log
Form B, $p = 1$	$b / [\bar{w}_d - \bar{w}_s + b]^2$	none (rational)